

Software Engineering Lab 2

Agile Backlog Creation & Sprint Simulation in Jira

Problem Statement: Problem Statement #09 – Campus & Academic Operations

System: Faculty Research Grant & Publication Tracker

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Section: A

Section 1: Agile Backlog Architecture (Epics & User Stories)

Epic	Story ID	Actor	User Story	Priority	Story Points
Epic 1: Grant Fund Management	US-01	Faculty Researcher	As a faculty researcher, I want to create and view my research grant fund details, so that I can keep track of my allocated research funding.	High	5
Epic 1: Grant Fund Management	US-02	Faculty Researcher	As a faculty researcher, I want to record approved expenses against my research grant, so that my grant spending is accurately tracked.	High	5
Epic 1: Grant Fund Management	US-03	Faculty Researcher	As a faculty researcher, I want the system to calculate my remaining grant balance after approved expenses, so that I know how much funding is still available.	Medium	3
Epic 2: Publication Management & Approval	US-04	Faculty Researcher	As a faculty researcher, I want to add and update my publication details, so that my research publications are maintained in one place.	High	3
Epic 2: Publication Management & Approval	US-05	Faculty Researcher	As a faculty researcher, I want to update indexing information and citation counts for my publications, so that I can track my publication performance.	Medium	3
Epic 2: Publication Management & Approval	US-06	Faculty Researcher	As a faculty researcher, I want to submit a publication to my co-author for review, so that the publication can be reviewed before approval.	High	5
Epic 2: Publication Management & Approval	US-07	Co-author	As a co-author, I want to approve or reject a submitted publication, so that I can provide my decision on the publication.	High	5
Epic 3: Research Monitoring & Analytics	US-08	Research Dean	As a Research Dean, I want to view faculty grant details and research activities, so that I can monitor ongoing research work.	Medium	5
Epic 3: Research	US-09	Research Dean	As a Research Dean, I	Medium	8

Monitoring & Analytics			want to view fund utilization reports and burn-up analytics, so that I can understand grant spending and remaining funds.		
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Total Backlog: 42 Story Points across 3 Epics and 9 User Stories.

Section 2: Sprint Execution & Board Simulation

Spaces / Faculty Research Grant and Publication Tracker

FRGAPT board

Summary

List

Active sprints

Backlog

Development

Forms

Timeline

Docs

Reports

Ask AI

Search work

AJ

Filter

Group

Work	Assignee	Reporter	Priority	Status
FRGAPT-1 Grant Fund Management	Unassigned	AJ AaroHi Jaiswal	Medium	Done
FRGAPT-2 Manage Research Grant Funds	Unassigned	AJ AaroHi Jaiswal	High	Done
FRGAPT-3 Record Grant Expenses	Unassigned	AJ AaroHi Jaiswal	High	Done
FRGAPT-4 Calculate Remaining Grant Balance	Unassigned	AJ AaroHi Jaiswal	Medium	Done
FRGAPT-5 Publication Management & Approval	Unassigned	AJ AaroHi Jaiswal	High	Done
FRGAPT-6 Manage Publication Records	Unassigned	AJ AaroHi Jaiswal	High	Done
FRGAPT-7 Track Publication Metrics	Unassigned	AJ AaroHi Jaiswal	Medium	Done
FRGAPT-9 Approve or Reject Publication	Unassigned	AJ AaroHi Jaiswal	High	Done
FRGAPT-8 Submit Publication for Review	Unassigned	AJ AaroHi Jaiswal	High	Done
FRGAPT-10 Research Monitoring & Analytics	Unassigned	AJ AaroHi Jaiswal	Medium	Done
FRGAPT-11 Monitor Research Activities	Unassigned	AJ AaroHi Jaiswal	Medium	Done
FRGAPT-12 View Fund Utilization Analytics	Unassigned	AJ AaroHi Jaiswal	Medium	Done

Grant Fund Management



✓ Description

Enable faculty researchers to manage research grant funds, record approved expenses, and monitor the remaining grant balance.

✓ Child work items

... [icon] +

100% Done

Work	Prio...	Story ...	Assi...	Status
☒ FRGAPT-2 Manage Research Grant Funds	^ H...	5	Un	Done ▾
☒ FRGAPT-3 Record Grant Expenses	^ H...	5	Un	Done ▾
☒ FRGAPT-4 Calculate Remaining Grant Balance	= M..	3	Un	Done ▾

Linked work items

Publication Management & Approval



✓ Description

Enable faculty researchers to maintain publication records and submit publications for co-author review and approval.

✓ Child work items

... [icon] -

100% Done

Work	Priority	Story ...	Assig...	Status
☒ FRGAPT-6 Manage Publication Records	^ Hi...	3	Una	Done ▾
☒ FRGAPT-7 Track Publication Metrics	= M...	3	Una	Done ▾
☒ FRGAPT-9 Approve or Reject Publication	^ Hi...	5	Una	Done ▾
☒ FRGAPT-8 Submit Publication for Review	^ Hi...	5	Una	Done ▾

Research Monitoring & Analytics



Description

Enable the Research Dean to monitor research activities and view grant utilization reports and burn-up analytics.

Child work items

... [icon] +

100% Done

Work	Priority	Story ...	Asslg...	Status
☒ FRGAPT-11 Monitor Research Activities	M...	5	Unassigned	Done
☒ FRGAPT-12 View Fund Utilization Analytics	M...	8	Unassigned	Done

Linked work items

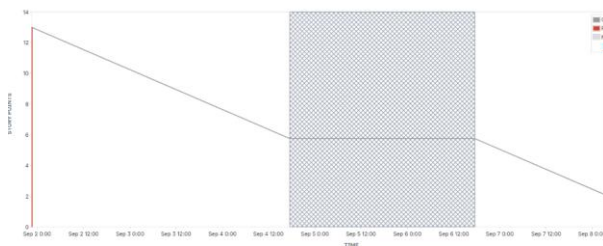
Section 3: Sprint Burndown & Metrics

Sprint-1

SPACES / Faculty Research Grant and Publication Tracker / FRGAPT-1 BOARD / Reports

Burndown Chart

FRGAPT Sprint 1 * Story Points *



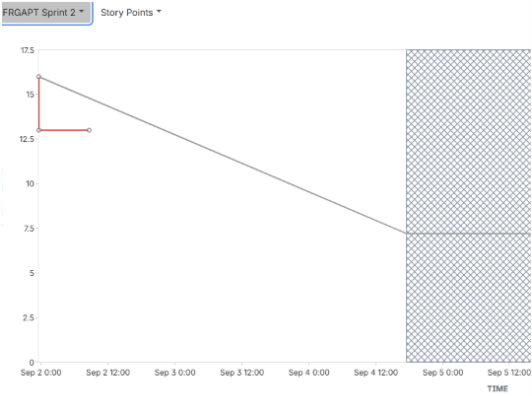
Spaces / Faculty Research Grant and Publication Tracker / FRGAPT board / Reports

Burndown Chart

How to read this chart

Date	Issue	Event type	Event Detail	Est.	Dec.	Remaining
01/Sep/26 10:36 PM	FRGAPT-2	Sprint start		5		
	FRGAPT-3			5		
	FRGAPT-4			3		
	Total					13
01/Sep/26 10:37 PM	FRGAPT-2	Burndown	Issue completed	5		8
	FRGAPT-3	Burndown	Issue completed	5		3
	FRGAPT-4	Burndown	Issue completed	3		0
01/Sep/26 10:38 PM	FRGAPT-2	Burndown	Issue reopened	5		5
	FRGAPT-3	Burndown	Issue reopened	5		10
	FRGAPT-4	Burndown	Issue reopened	3		13
01/Sep/26 10:39 PM	FRGAPT-2	Burndown	Issue completed	5		8
	FRGAPT-4	Burndown	Issue completed	3		5
	FRGAPT-3	Burndown	Issue completed	5		0
	FRGAPT-2	Sprint ended by Aarohi Jalilwal				0
	FRGAPT-3					0
	FRGAPT-4					0
	Total					0

Sprint-2



Date	Issue	Event Type	Event Detail	Story Points		
				Inc.	Dec.	Remaining
01/Sep/26 11:32 PM	FRGAPT-6	Sprint start		3		
	FRGAPT-7			3		
	FRGAPT-8			5		
	FRGAPT-9			5		
	Total					16
01/Sep/26 11:35 PM	FRGAPT-6	Burndown	Issue completed		3	13
02/Sep/26 8:33 AM	FRGAPT-6	Sprint ended by Aarohi Jaiswal				0
	FRGAPT-7					3
	FRGAPT-8					5
	FRGAPT-9					5
	Total					13

Sprint 3



Date	Issue	Event Type	Event Detail	Story Points		
				Inc.	Dec.	Remaining
03/Sep/26 1:56 PM	Total	Sprint start				0
	FRGAPT-11	Scope change	Issue added to sprint	5		5
	FRGAPT-12	Scope change	Issue added to sprint	8		13
04/Sep/26 9:01 PM	FRGAPT-11	Burndown	Issue completed		5	8
04/Sep/26 10:30 PM	FRGAPT-12	Burndown	Issue completed		8	0
	FRGAPT-11	Sprint ended by Aarohi Jaiswal				0
	FRGAPT-12					0
	Total					0

Analysis

The burndown charts compare the ideal guideline with the remaining story points during each sprint. Sprint 1 contains 13 story points, Sprint 2 contains 16 story points, and Sprint 3 contains 13 story points, completing the total backlog of 42 story points. The actual line decreases as stories are completed and moved to Done. The charts show that the planned work was completed within each simulated sprint.

Section 4: Reflection & Analysis

Question 1: Did your estimations reflect the actual effort?

The Fibonacci scale was used to estimate story effort. Smaller stories such as managing publication records were assigned 3 points, while workflows involving multiple steps or dependencies were assigned 5 points. The analytics story was assigned 8 points because it combines grant data, expenses, balances, spending trends, and report generation. The Jira simulation showed that the estimated stories could be completed within the planned sprints, indicating that the estimates were reasonable for the simulated workload.

Question 2: Was your backlog well-prioritized?

The backlog was prioritized using the Lab 1 functional-requirement priorities and feature dependencies. Core grant management and publication submission/approval receive High priority, while publication metrics, balance calculation, research monitoring, and analytics are Medium priority where appropriate. This provides a logical development sequence.

Question 3: How did your simulated sprint align with your plan?

The work was divided into three simulated sprints based on the available story points. Sprint 1 focused on grant fund management with 13 story points, Sprint 2 focused on publication management and approval with 16 story points, and Sprint 3 focused on research monitoring and analytics with 13 story points. The Jira screenshots show that the planned stories were moved through the sprint workflow and completed successfully.

Question 4: What insights did the burndown chart give about your team's capacity?

The burndown charts show how the remaining story points decreased as stories were completed. Sprint 1 reduced its workload from 13 to 0 points, Sprint 2 from 16 to 0 points, and Sprint 3 from 13 to 0 points. This indicates that the simulated team was able to complete the committed work in each sprint and that the sprint workload was manageable.